

Inverse By Row Elementary operation

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$\text{determinant} = -1$$

$$[A : I]$$

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 1 & 1 & -3 & 1 & -2 & 1 & 0 \\ 1 & 0 & 8 & 1 & 0 & 0 & 1 \end{bmatrix} \quad R_2 - R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 1 & 1 & -3 & 1 & -2 & 1 & 0 \\ 0 & -2 & 5 & 1 & -1 & 0 & 1 \end{bmatrix} \quad R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & -3 & 1 & -2 & 1 & 0 \\ 0 & 0 & -1 & 1 & -5 & 2 & 1 \end{bmatrix} \quad R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & -3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -5 & 2 & -1 \end{bmatrix} \rightarrow R_3$$

Operations

$$R_0 \leftrightarrow R_m$$

$$R_0 \leftrightarrow R_m$$

$$a R_0 \quad \left\{ \begin{array}{l} \text{interchange} \\ \text{a row by} \\ \text{2nd row} \end{array} \right.$$

$$1 R_0 \quad a \neq 0$$

$$No \text{ use} \quad \left\{ \begin{array}{l} R_n \times R_m \\ \text{is not a} \end{array} \right.$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 7 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] R_1 - 2R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \begin{array}{l} R_1 - 7R_3 \\ R_2 + 3R_3 \end{array}$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

$$A A^{-1} = I$$

solved

$$Q: \left[\begin{array}{ccc|c} 1 & 2 & 3 & \\ 2 & 3 & 4 & R_2 - 2R_1 \\ 3 & 5 & 7 & \end{array} \right]$$

$$A = \left[\begin{array}{ccc|c} 1 & 2 & 3 & \\ 0 & -1 & -2 & R_3 - 3R_1 \\ 3 & 5 & 7 & \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & \\ 0 & -1 & -2 & R_3 - R_2 \\ 0 & -1 & -2 & \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & \\ 0 & 0 & -2 & \\ 0 & 0 & 0 & \end{array} \right]$$

~~Ans~~

Rank of matrix

↳ Number of non-zero ^{rows in} matrix after ~~with~~ ^{row} ~~norm~~

1 entry $\rightarrow$ leading entry $\rightarrow$

↳ always one ^{row}

make upper triangular matrix of given matrix.

↳ less than order
rank

↳ no rows

Q1

check
do it
apply the rule
do solve system
of 2 equations
or to analyze
vector space

→ Gauss, Jordan method

II Solution of System of linear Equations.

→ Gauss elimination method

→ $Ax = B \rightarrow \text{constant}$
 $\swarrow \quad \searrow$
 coefficient variable matrix

matrix must be square matrix.

Q: $x - y + 2z = 3$

$2x + 2y + 3z = 5$

$3x - 4y - 5z = -13$

Sol

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 2 & 3 \\ 3 & -4 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ -13 \end{bmatrix}$$

A|B

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 2 & 2 & 3 & 5 \\ 3 & -4 & -5 & -13 \end{array} \right] \quad 3R_2 - R_3$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 2 & 1 & 2 \\ 3 & -11 & -5 & -12 \end{array} \right] \quad R_3 - 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 2 & 1 & 2 \\ 0 & -13 & -11 & -15 \end{array} \right] \quad \begin{array}{l} R_3 + R_2 \\ R_3 - 13R_2 \end{array}$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & -32 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -64 \end{bmatrix}$$

$$x - y + 2z = 3 \quad \rightarrow 1$$

$$3y + z = 2 \quad \rightarrow 2$$

$$-32z = -64 \quad \rightarrow 3$$

$$z = -2$$

$$x = -1$$

$$y = 0$$

$$[A : B] = C$$

if $R(B) = C$ & no row zero in C then consistent solvable

Consistency of System of linear Equation
 ↳ more than one value

a) Consistent: if rank of matrix(A) = Rank of matrix(C)

↳ Unique Solution:
 $\text{Rank}(A) = \text{Rank}(C) = n \rightarrow$ no. of variables

↳ Infinite Solution:
 $\text{Rank}(A) = \text{Rank}(C) < n$

b) In consistent: $\text{Rank}(A) \neq \text{Rank}(C)$

$$\text{Q: } 5x + 3y + 7z = 4$$

$$3x + 26y + 2z = 9$$

$$7x + 2y + 10z = 5$$

$$\Rightarrow \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix}$$

Sol:

$$A \cdot B = C$$

$$\left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 5R_2 - 3R_1$$

$$\left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 7 & 2 & 10 & 5 \end{array} \right]$$

$$5R_3 - 7R_1$$

$$\left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 0 & 121 & -11 & 33 \\ 0 & -11 & 1 & -3 \end{array} \right]$$

$$11R_3 + R_2$$

Q. 2

$$\begin{bmatrix} 5 & 3 & 7 & | & 4 \\ 0 & 121 & -11 & | & 33 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Rank of matrix = 2

Infinite solutions

$$\begin{bmatrix} 5 & 3 & 7 \\ 0 & 121 & -11 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 33 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 5x + 3y + 7z &= 4 \rightarrow 1 \\ + 121y - 11z &= 33 \rightarrow 2 \\ 0 &= 0 \end{aligned}$$

let $z = k$

$$y = \frac{33 + 11k}{121}$$

$$x = \frac{5x + 3\left(\frac{33 + 11k}{121}\right) + 7k = 4}{5}$$

$$x = 4 - 7k - 3\left(\frac{33 + 11k}{121}\right)$$

$$x = -9/11$$

$$y = 4/11$$

$$z = 1$$

$$5(-9/11) + 3(4/11) + 7 = 4$$

$$3(-9/11) + 2(4/11) + 2 = 2$$

$$7(-9/11) + 2(4/11) + 10 = 5$$

$$\begin{aligned}
 \text{Q1. } & x_1 + 2x_2 + x_3 = 2 \\
 & 3x_1 + x_2 - 2x_3 = 1 \\
 & 4x_1 - 3x_2 - x_3 = 3 \\
 & 2x_1 + 4x_2 + 2x_3 = 4
 \end{aligned}$$

Sol.

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & -2 \\ 4 & -1 & -1 \\ 2 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & | & 2 \\ 3 & 1 & -2 & | & 1 \\ 4 & -1 & -1 & | & 3 \\ 2 & 4 & 2 & | & 4 \end{bmatrix} \quad 2R_4 - R_3$$

$$\begin{bmatrix} 1 & 2 & 1 & | & 2 \\ 3 & 1 & -2 & | & 1 \\ 4 & -1 & -1 & | & 3 \\ 0 & 9 & 5 & | & 5 \end{bmatrix} \quad \cancel{3R_4 - R_3} \quad R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & | & 2 \\ 3 & 1 & -2 & | & 1 \\ 0 & -7 & -5 & | & -5 \\ 0 & 9 & 5 & | & 5 \end{bmatrix} \quad R_2 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 2 \\ 0 & -5 & -5 & 1 & -5 \\ 0 & -9 & -5 & 1 & -5 \\ 0 & 9 & 5 & 1 & 5 \end{bmatrix} \quad R_2 + R_3$$

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 2 \\ 0 & -5 & -5 & 1 & -5 \\ 0 & -9 & -5 & 1 & -5 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad 2R_3 - 9R_4$$

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 2 \\ 0 & -5 & -5 & 1 & -5 \\ 0 & 0 & -19 & 1 & -28 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

unique solution.
Rank (3)

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & -5 \\ 0 & 0 & -19 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \\ -28 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 2x - 10y - 28z &= 2 \quad (1) \\ +25y + 28z &= -5 \quad (2) \\ +532z &= -28 \quad (3) \end{aligned}$$

For what value of λ and 21 ,
so that the equations

$$2x + 3y + 5z = 9$$

$$7x + 3y - 2z = 8$$

$$2x + 3y + \lambda z = 11$$

have

- ① No Solution. $\rightarrow$ Rank $\neq$ Rank(B) $\rightarrow$ not consistent
 ② a Unique Solution $\rightarrow$ Rank $=$ Rank(B) $\rightarrow$ consistent
 ③ an infinite number of Solution $\rightarrow$ Rank(A) = Rank(B) $< n$

$$\begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \\ 11 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 2 & 3 & \lambda & 11 \end{array} \right] R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 0 & 0 & \lambda - 5 & 11 - 9 \end{array} \right] 2R_2 - 7R_1$$

$$\left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 0 & -15 & -39 & -47 \\ 0 & 0 & \lambda - 5 & 11 - 9 \end{array} \right] \cancel{7R_1} \quad \cancel{7R_1}$$

① for no solution

$$\lambda = 5, \mu \neq 9$$

② for unique solution

$$\lambda \neq 5, \mu \neq 9$$

③ Infinite solution

$$\lambda = 5, \mu = 9$$

Q2: $x + y + z = 6$

$$2x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 2 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix} \begin{array}{l} \\ R_2 - R_1 \\ R_3 - R_1 \end{array}$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{bmatrix} \begin{array}{l} \\ \\ R_3 - R_2 \end{array}$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{bmatrix}$$

no sol ① $\lambda = 3, \mu \neq 10$

Unique ② $\lambda \neq 3, \mu \neq 10$

Infinite: ③ $\lambda = 3, \mu = 10$

14 order matrix

For what value of k the equation

$$x + y + z = 1$$

$$2x + y + 4z = k$$

$$4x + y + 10z = k^2$$

has a solution.

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 4 \\ 4 & 1 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ k \\ k^2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 4 & 1 & k \\ 4 & 1 & 10 & 1 & k^2 \end{bmatrix} \quad R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 4 & 1 & k \\ 0 & -1 & 2 & -1 & k^2 - 2k \end{bmatrix} \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & -1 & k-2 \\ 0 & -1 & 2 & -1 & k^2-2k \end{bmatrix} \quad R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & -1 & k-2 \\ 0 & 0 & 0 & 0 & k^2-k-2 \end{bmatrix}$$

k^2-k-2



tuples $\rightarrow$ 3 values

Linear dependence & independence of Vectors (matrices)

Vectors (matrices) $x_1, x_2, x_3, \dots, x_n$ are said to be dependent if,

(i) all vectors (row or column matrices) are of the same order

(ii) for scalar $\lambda_1, \lambda_2, \dots, \lambda_n$ (not all zero), exist such that

$$\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \dots + \lambda_n x_n = 0$$

Otherwise they are linearly independent.

eg

Examine the following vectors for linear dependence and find the relation if it exist.

$$x_1 = (1, 2, 4), x_2 = (2, -1, 3), x_3 = (0, 1, 2) \text{ and } x_4 = (-3, 7, 2)$$

Sol

$\rightarrow$ System of equations based on King's rule
 $\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \lambda_4 x_4 = 0$ is King's column

$$\lambda_1(1, 2, 4) + \lambda_2(2, -1, 3) + \lambda_3(0, 1, 2) + \lambda_4(-3, 7, 2) = 0$$

$$\lambda_1 + 2\lambda_2 + 0\lambda_3 - 3\lambda_4 = 0$$

$$2\lambda_1 - \lambda_2 + \lambda_3 + 7\lambda_4 = 0$$

$$4\lambda_1 + 3\lambda_2 + 2\lambda_3 + 2\lambda_4 = 0$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 2 & -1 & 1 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix} \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 4 & 3 & 2 & 2 \end{bmatrix} \quad R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & -5 & 2 & 14 \end{bmatrix} \quad R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \begin{matrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} \quad \text{so}$$

$$\lambda_1 + 2\lambda_2 - 3\lambda_4 = 0 \rightarrow \textcircled{a}$$

$$-5\lambda_2 + \lambda_3 + 13\lambda_4 = 0 \rightarrow \textcircled{b}$$

$$\lambda_3 + \lambda_4 = 0 \rightarrow \textcircled{c}$$

from $\textcircled{c}$

$$\lambda_3 = -\lambda_4$$

$$\text{let } \lambda_3 = t$$

$$\lambda_4 = -t$$

$$\lambda_2 = -\frac{12}{5}t$$

$$\lambda_1 = 9/5$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 9/5 \\ -12/5 \\ t \\ -t \end{bmatrix}$$

$$\frac{24}{15}$$

$$9/5(1, 2, 4) - 12/5(2, -1, 3) + (0, 1, 2)(3, 2, 2)$$

$$\left(\frac{9}{5} + \frac{12}{5} + 3\right) = 0 \quad = 0$$

$$\frac{9}{5} = \frac{24}{5} + 0 + 3$$

$$\frac{9 - 24 + 0 - 15}{5} = \frac{15 - 15}{5} \quad 9/5 = 0$$

$$2) \quad 2 \times \frac{9}{5} + \frac{12}{5} + 1 - 7 =$$

$$\frac{18}{5} + \frac{12}{5} + \frac{1}{1} - \frac{7}{1}$$

$$\frac{18+12+5-35}{5} = \frac{0}{5} = 0 = 0$$

$$3) \quad 4 \times \frac{9}{5} - \frac{36}{5} + 2 - 2 =$$

$$\frac{36}{5} - \frac{36}{5} + 2 - 2 = 0$$

Relation is dependent

Linear dependence & independence by

Rank Method

① The rank of the matrix of the given matrix is equal to the number of vectors, then the given vectors are linearly independent. $\rightarrow$ Rank

② If rank is less than the given vectors are linearly dependent. $\rightarrow$ less not greater

$\Rightarrow$ Show that the following vectors are linearly dependent by using rank method

$$X = [1, 2, -3, 4], Y = [3, -1, 2, 1], Z = [1, -5, 8, -7]$$

sol:

ids row R form main likhna.

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 2 & 1 \\ 1 & -5 & 8 & -7 \end{bmatrix} \quad R_2 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 1 & -5 & 8 & -7 \end{bmatrix} \quad R_3 - R_1$$

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 0 & -7 & 11 & -11 \end{bmatrix} \quad R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 0 & 0 & 13 & -15 \end{bmatrix}$$

Cayley-Hamilton Theorem
 ↳ Characteristic equation $|A - \lambda I| = 0$
 ↳ value ki jagah matrices put karate
 Or sum zero ho
 ↳ matrix sum with matrix which is identity matrix & multiply with matrix

$$\lambda^2 - a\lambda + b = 0$$

$$\hookrightarrow \text{matrix} = a\lambda + bI = 0$$

Q:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$|A - \lambda I| = 0$$

26/4/21

$$\begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} - \lambda \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & -1-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} & (1-\lambda)(-1-\lambda) - 2 \times 2 = 0 \\ & -1 - \lambda + \lambda + \lambda^2 - 4 = 0 \\ & \lambda^2 - 5 = 0 \end{aligned} \quad \begin{array}{r} 2 \overline{) 1} \\ 2 \overline{) 5} \text{ rd} \\ 2 \overline{) 2} - 1 \end{array}$$

For C.H.T.

$$\lambda^2 - 5 = 0$$

$$\lambda^2 - 5I = 0$$

$$\begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} - 5 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$\begin{aligned} & (1-5) - (2 \cdot 2) - (2 \cdot 2) - (4-5) = 0 \\ & -4 - 4 - 4 - 5 = 0 \end{aligned}$$

$$\begin{vmatrix} 5 & 0 \\ 0 & 5 \end{vmatrix} - 5 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 0$$

$$= \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

hence prove

for inverse

$$Q. \begin{vmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{vmatrix} - \lambda \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 4-\lambda & 3 & 1 \\ 2 & 1-\lambda & -2 \\ 1 & 2 & 1-\lambda \end{vmatrix}$$

$$(4-\lambda)(1-\lambda)(1-\lambda) - 3((1-\lambda)(1-\lambda))$$

9)

$$(4-\lambda) [(1-\lambda)(1-\lambda) - (-2)(2)] - 3 [(2)(1-\lambda) - (-2)(1)] +$$

$$1 [(2)(2) - (1)(1-\lambda)]$$

$$(4-\lambda) [1-\lambda-\lambda+\lambda^2+4] - 3 [2-2\lambda+2] + [4-1+\lambda]$$

$$(4-\lambda) [\lambda^2-2\lambda+5] - 3 [2-2\lambda+2] + [4-1+\lambda]$$

$$4\lambda^2-8\lambda+20-\lambda^3+2\lambda^2-5\lambda-6+6\lambda-6+4-1+\lambda$$

$$-\lambda^3+4\lambda^2+2\lambda^2-8\lambda-5\lambda+6\lambda+\lambda+20-6-6+4-1$$

$$-\lambda^3+6\lambda^2-6\lambda+11=0$$

~~$$-\lambda^3+6\lambda^2-6\lambda+11=0$$~~

~~$$-\lambda^2(\lambda-6)-6(\lambda-11/6)=0$$~~

~~12~~

multiply

$$\lambda^3-6\lambda^2+6\lambda-11=0$$

$$\lambda^3-6\lambda^2+6\lambda-11=0$$

$$\lambda^3-6\lambda^2+6\lambda+11$$

$$A^3 - 6A^2 + 6A - 11I$$

for A^3
1st find A^2

$$A^2 = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 16 + 6 + 1 & 12 + 3 + 2 & 4 - 6 + 1 \\ 8 + 2 - 2 & 6 + 1 - 4 & 2 - 2 - 2 \\ 4 + 4 + 1 & 3 + 2 + 2 & 1 - 4 + 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 23 & 17 & -1 \\ 8 & 3 & -2 \\ 9 & 7 & -2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 23 & 17 & -1 \\ 8 & 3 & -2 \\ 9 & 7 & -2 \end{bmatrix} \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$A^3 = \begin{vmatrix} 92+34-1 & 69+17-2 & 23-34-1 \\ 32+6-2 & 24+3-4 & 8-6-2 \\ 36+14-2 & 27+7-4 & 9-14-2 \end{vmatrix}$$

$$A^3 = \begin{vmatrix} 125 & 84 & -12 \\ 36 & 23 & 0 \\ 48 & 30 & -7 \end{vmatrix}$$

$$A^3 - 6A^2 + 6A - 11I = 0$$

$$A^3 - 6A^2 + 6A = 11I$$

$$\begin{vmatrix} 125 & 84 & -12 \\ 36 & 23 & 0 \\ 48 & 30 & -7 \end{vmatrix} - 6 \begin{vmatrix} 23 & 17 & -1 \\ 8 & 3 & -2 \\ 9 & 7 & -2 \end{vmatrix} + 6 \begin{vmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{vmatrix} = 11I$$

$$\begin{vmatrix} 125 & 84 & -12 \\ 36 & 23 & 0 \\ 48 & 30 & -7 \end{vmatrix} - \begin{vmatrix} 138 & -102 & +6 \\ -48 & -18 & 12 \\ -48 & -42 & +12 \end{vmatrix} + \begin{vmatrix} 24 & 18 & 6 \\ 12 & 6 & -12 \\ 6 & 12 & 6 \end{vmatrix} = 11I$$

$$\begin{vmatrix} 125 & 84 & -12 \\ 36 & 23 & 0 \\ 48 & 30 & -7 \end{vmatrix} - \begin{vmatrix} 138 & -102 & +6 \\ -48 & -18 & 12 \\ -48 & -42 & 12 \end{vmatrix} + \begin{vmatrix} 24 & 18 & 6 \\ 12 & 6 & -12 \\ 6 & 12 & 6 \end{vmatrix} = \begin{vmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{vmatrix}$$

first $A^3 - 6A^2$

$$= \begin{bmatrix} 125 - 138 & 84 - 102 & -12 - (6) \\ 36 - 48 & 23 - 18 & 0 + 12 \\ 48 - 48 & 30 - 42 & -7 + 12 \end{bmatrix} = \begin{bmatrix} -13 & -18 & -6 \\ -12 & 5 & 12 \\ 0 & -12 & 5 \end{bmatrix}$$

Add $+6A$

$$\begin{bmatrix} -13 & -18 & -6 \\ -12 & 5 & 12 \\ 0 & -12 & 5 \end{bmatrix} + \begin{bmatrix} 24 & 18 & 6 \\ 12 & 6 & -12 \\ 6 & 12 & 6 \end{bmatrix} = \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix}$$

Sub $11I$

$$\begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} - \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

0

= 0

Proved

$$\text{Q1 } A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

Sol:

$$|A - \lambda I| = 0$$

$$= \begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix}$$

$$= (1-\lambda) \begin{vmatrix} 2-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} - 0 + 1 \begin{vmatrix} 1 & 2-\lambda \\ 2 & 2 \end{vmatrix}$$

$$= (1-\lambda) [(2-\lambda)(3-\lambda) - 2] - 0 + 1(2) - 2(2-\lambda)$$

$$= (1-\lambda) [\lambda^2 - 5\lambda + 4] - 0 + 2\lambda - 2$$

$$= \lambda^3 - 6\lambda^2 + 11\lambda - 6$$

$$= (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$$

$$\lambda = 1, 2, 3$$

Eigen values

For Eigenvector
 $\lambda = 1$

$$= \begin{vmatrix} 1-1 & 0 & -1 \\ 1 & 2-1 & 1 \\ 2 & 2 & 3-1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{vmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$-z = 0 \Rightarrow z = 0$$

$$x + y + z = 0 \Rightarrow x + y = 0 \Rightarrow x = -y$$

$$2x + 2y + 2z = 0$$

~~2=0~~

suppose $x = -t$

$$y = t$$

$$z = 0$$

$$\Rightarrow \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$\lambda = 2$$

$$= \begin{vmatrix} 1-2 & 0 & -1 \\ 1 & 2-2 & 1 \\ 2 & 2 & 3-2 \end{vmatrix} = \begin{vmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{vmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$-x - z = 0 \Rightarrow x = -z$$

~~2=0~~

$$x + z = 0 \Rightarrow -z + z = 0 \Rightarrow 0$$

$$2x + 2y + z = 0$$

$$2(-z) + 2y + z = 0$$

$$-2z + 2y + z = 0 \Rightarrow -z + 2y = 0 \Rightarrow z = 2y$$

$$y = z/2$$

suppose $x = -2$

$$z = 2$$

$$y = 1$$

$$\begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$$

$$\lambda = 3$$

$$\begin{bmatrix} 1-3 & 0 & -1 \\ 1 & 2-3 & 1 \\ 2 & 2 & 3-3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -2 & 0 & -1 & x \\ 1 & -1 & 1 & y \\ 2 & 2 & 0 & z \end{array} \right] = 0$$

$$-2x - z = 0 \Rightarrow z = -2x$$

$$x - y + z = 0 \Rightarrow x - y - 2x = 0$$

$$-x - y = 0 \Rightarrow x = -y/y = -x$$

$$2x + 2y = 0$$

$$x + x - 2x = 0$$

$$0 = 0$$

$$y = -x$$

$$z = -2x$$

$$\text{suppose } x = 1$$

$$\begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$$

$$P = \begin{bmatrix} -1 & -2 & 1 \\ 1 & 1 & -1 \\ 0 & 2 & -2 \end{bmatrix}$$

$$P^{-1} = \left[\begin{array}{ccc|ccc} -1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & -2 & 0 & 0 & 1 \end{array} \right] R_1 \leftrightarrow R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ -1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 2 & -2 & 0 & 0 & 1 \end{array} \right] R_2 + R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 & 0 & 1 \end{array} \right] R_2 \rightarrow -R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 2 & -2 & 0 & 0 & 1 \end{array} \right] R_3 - 2R_2$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & -2 & 2 & 2 & 1 \end{array} \right] R_3 \left| \begin{array}{l} -2 \\ -2 \end{array} \right.$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & -1 & 0 \\ 0 & 0 & 1 & -1 & -1 & -1/2 \end{array} \right] R_1 + R_3$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 & -1 & 0 & -1/2 \\ 0 & 1 & 0 & 1 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 & -1 & -1 & -1/2 \end{bmatrix} \quad R_1 - R_2$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & -1/2 \\ 0 & 1 & 0 & 1 & -1 & -1 & 0 \\ 0 & 0 & 1 & 1 & -1 & -1 & -1/2 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} 0 & 1 & -1/2 \\ -1 & -1 & 0 \\ -1 & -1 & -1/2 \end{bmatrix}$$

Now

$$D = P^{-1} A P$$

$$D = \begin{bmatrix} 0 & 1 & -1/2 \\ -1 & -1 & 0 \\ -1 & -1 & -1/2 \end{bmatrix} \begin{bmatrix} -1 & -2 & 1 \\ 1 & 1 & -1 \\ 0 & 2 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

proved

LINEAR ALGEBRA & GEOMETRY FORMULAS MATRIX

Row matrix

1 $\rightarrow$ Row

any no: of Column.

Column Matrix

1 $\rightarrow$ column

any no: of row

Null / zero matrix

All elements zero

Square matrix

No: of rows = no: of column

Diagonal matrix

all non-diagonal element zero

(Square matrix)

Scalar matrix

all diagonal element equal to be a scalar, say k
 $A = [a_{ij}]_{n \times n} \Rightarrow$ if $a_{ij} = \begin{cases} 0 & \rightarrow \text{when } i \neq j \\ k & \rightarrow \text{when } i = j \end{cases}$

Unit / Identity matrix

diagonal element $= 1$

non diagonal $= \text{zero}$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Symmetric matrix

$$A^t = A \text{ or } a_{ij} = a_{ji}$$

Skew Symmetric matrix

$$1) A^t = -A \text{ or } a_{ij} = -a_{ji}$$

2) All diagonal element are zero

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

Triangular matrix (Echelon form)

leading diagonal $= \text{zero}$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

Transpose of a matrix

$A = A^t \Rightarrow$ transpose $\rightarrow$ Row changes into column

Orthogonal matrix

$$A \cdot A^t = I$$

Conjugate of matrix

$$A = \bar{A}$$

$i \rightarrow \text{sign change}$

Matrix

$$A^o$$

$$A^o = (\bar{A})^t$$

Unitary matrix

$$A^0 A = I \Rightarrow A^0 = (\bar{A})^t \Rightarrow \text{transpose of conjugate}$$

Hermitean matrix

$$a_{ij} = \bar{a}_{ji}$$

$$A = (\bar{A})^t \Rightarrow \text{conjugate transpose of } A$$

Skew H.M

$$a_{ij} = -a_{ji}$$

Idempotent matrix

$$A^2 = A$$

Periodic matrix

$$A^{K+1} = A \quad \therefore K=1 \text{ then } A^2 = A$$

Nil Potent

$$A^K = 0 \quad \therefore K = +ve \text{ integer}$$

$$A^2 = 0$$

Involuntary matrix

$$A^2 = \text{Identity matrix.}$$

Equal matrix.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \Rightarrow A=B$$

Singular matrix

$$\text{Determinant} = 0$$

Upper triangular matrix

$$A = \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$$

Lower triangular matrix

$$A = \begin{pmatrix} a & 0 & 0 \\ b & d & 0 \\ c & e & f \end{pmatrix}$$

Equality of two matrix

$$A = (a_{ij})_{m \times n} \quad \& \quad B = (b_{ij})_{m \times n} \Rightarrow a_{ij} = b_{ij}$$

Scalar multiple of a matrix

$$kA = (ka_{ij})_{m \times n}$$

Inverse By Rows Elementary Operation

Determinant $\neq 0$

$$[A : I]$$

$$\text{then } AA^{-1} = I$$

$$R_n \pm R_m$$

$$R_n \leftrightarrow R_m$$

$$aR_n$$

$$\frac{1}{a}R_n \rightarrow a \neq 0$$

$$R_n \times R_m$$

$$R_n \pm a \quad \left. \begin{array}{l} R_n \times R_m \\ R_n \pm a \end{array} \right\} \text{no use}$$

Rank of Matrix

Not : of non-zero rows after Echelon form.

Leading entry $\rightarrow$ always (1)

Solution Of System Of Linear Equation

$AX = B \rightarrow \text{constant}$
 $\swarrow$ coefficient $\searrow$ variable
matrix must be square matrix

Consistency of System of Linear Equation

$$[A : B] = C$$

Consistent : $\text{Rank}(A) = \text{Rank}(C)$

Unique Solution : $R(A) = R(C) = \text{no. of variables}$

Infinite Solution : $R(A) = R(C) < n$

In consistent :

$$\text{Rank}(A) \neq \text{Rank}(C)$$

Eigen Values & Eigen Vectors

$\hookrightarrow$ values of scalar

~~$$[A - \lambda I]x = 0$$~~

$$[A - \lambda I] = 0 \rightarrow \text{For values}$$

$\hookrightarrow$ characteristic equation

For vectors

$$[A - \lambda I]x = 0$$

Linear dependence & Independence of Vector

- 1) all vectors (Rows or Column matrices) are of same order
- 2) $\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \dots + \lambda_n x_n = 0$

Linear dependent

otherwise independent.

Cayley Hamilton Theorem

$$|A - \lambda I| = 0$$

↳ Sum = 0

values ki jage matrix put karain

$$\begin{cases} \lambda^2 = a\lambda + b = 0 \\ \text{matrix} = a\lambda + bI = 0 \end{cases}$$

Diagonalization

$$D = P^{-1}AP \rightarrow \text{matrix of the eigen vectors}$$

3D Co-ordinate System

Polar $\rightarrow$ 2D cartision

$$\begin{aligned} \text{Polar} = r &= \sqrt{x^2 + y^2} \\ \theta &= \tan^{-1} y/x \end{aligned}$$

$$(x, y) = r(\cos\theta + i\sin\theta)$$

$$\text{Polar} = (r, \theta)$$

Cylindrical

$$(x, y, z) = (r, \theta, z)$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

Conditions

$$\theta = \tan^{-1}(y/x)$$

- If $x > 0$, $y > 0$
 $\theta = \text{same}$

- If $x < 0$, and $y > 0$
 $\theta = \pi$

- If $x < 0$ and $y < 0$
 $\pi - \theta$

- If $x > 0$ and $y < 0$
 $-\theta$

Spherical to Cartesian

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$\therefore \phi = \cos^{-1}(z/\rho)$$